

Roman Numerals

Problem Description

The Romans wrote their numbers using letters; specifically the letters 'I' meaning '1', 'V' meaning '5', 'X' meaning '10', 'L' meaning '50', 'C' meaning '100', 'D' meaning '500', and 'M' meaning '1000'. There were certain rules that the numerals followed which should be observed.

The symbols 'I', 'X', 'C', and 'M' can be repeated at most 3 times in a row. The symbols 'V', 'L', and 'D' can never be repeated. The '1' symbols ('I', 'X', and 'C') can only be subtracted from the 2 next highest values ('IV' and 'IX', 'XL' and 'XC', 'CD' and 'CM'). Only one subtraction can be made per numeral ('XC' is allowed, 'XXC' is not). The '5' symbols ('V', 'L', and 'D') can never be subtracted.

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1 → I
2 → II
3 → III
4 → IV
5 → V
9 → IX
21 → XXI
50 → L
100 → C
500 → D
1000 → M
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Part I – Convert Arabic to Roman Numeral

We would like to be able to convert Arabic numbers into their Roman numeral equivalents. We just need some kind of program that can accept a numeric input and output the Roman numeral for the input number.

Part II - Convert Roman Numeral to Arabic

We would like to be able to convert Roman numerals to Arabic numbers. We just need some kind of program that can accept a Roman numeral as input and output the Arabic number for the input.